



EVIDENCE GUIDE

# Box3D Metal Compatibility and Validation

The exact GPU-resident surface, explicit CPU boundary, differential evidence, and completed acceptance matrix.



# Compatibility contract

The reference is unmodified Box3D commit 47d7f7cc7e091142c08d11dc7d2e493c5d34f536. Metal is opt-in and tolerance-equivalent, not bit-identical across platforms.

| GPU-resident surface | Modes                                                           |
|----------------------|-----------------------------------------------------------------|
| Fused integration    | Velocity and position across all substeps                       |
| Convex contacts      | Normal, friction, tangent velocity, twist, rolling, restitution |
| Mesh contacts        | Scalar multi-manifold solve and restitution                     |
| Distance joints      | Rigid, spring, limit, motor, static bodies                      |
| Parallel joints      | Soft alignment, torque limiting, static bodies                  |
| Overflow             | Serial deterministic contact and mixed supported-joint order    |
| Pair traversal       | Experimental exact-order raw dynamic-tree candidates            |

## Explicit CPU boundary

- Broad-phase tree mutation, pair filtering, narrow phase, and manifolds
- Contact and joint preparation
- Events, islands, sleeping, and CCD
- Recording, queries, topology mutation, and public API calls
- Filter, motor, prismatic, revolute, spherical, weld, and wheel solving
- Any joint requesting reaction-threshold events

Unsupported constraints keep the reference CPU solve for that step. The position-only Metal stage may still run if its threshold is met.

## Differential evidence

Matching CPU and Metal worlds run for multiple steps/substeps. Transforms and linear/angular velocities are compared after each scenario. Values below are observed maxima, not universal bounds.

| Scenario                | Transform         | Velocity         |
|-------------------------|-------------------|------------------|
| Convex friction         | 4.77e-7 transform | 3.98e-6 velocity |
| Mesh contacts           | 4.77e-7           | 3.60e-6          |
| Distance modes/overflow | 1.07e-6           | 8.27e-5          |
| Mixed distance/parallel | 7.45e-9           | 1.79e-7          |
| Mixed joint overflow    | 1.82e-6           | 3.29e-4          |
| Static-body joints      | 4.47e-8           | -                |

## Acceptance matrix

Debug and Release full suites; CPU-only Release; double-precision Metal differential tests; AddressSanitizer and UndefinedBehaviorSanitizer; warning-as-error build; shared dylib demo; CMake install audit; and clean-clone patch reconstruction.

Failure semantics. Initialization failure returns false and leaves the world usable. Unsupported work increments fallback counters. Disabling Metal releases resources without destroying the world.